

LECTURE SEVEN (153 248)

SINUSOIDAL MODEL

y_t : observed time series

$$y_t = \beta_0 + \beta_1 \cos(2\pi ft) + \beta_2 \sin(2\pi ft) + \varepsilon_t$$

unknown parameters

$\beta_0, \beta_1, \beta_2, f, \sigma$

$\varepsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$
error
or
noise
term

$$y_t = \beta_0 + \beta_1 t + \beta_2 (t - c)_+ + \varepsilon_t$$

$\beta_0, \beta_1, \beta_2, c, \sigma$

$$y = X_f \beta + \varepsilon$$

$$X_f = \begin{bmatrix} 1 & \cos(2\pi ft) & \sin(2\pi ft) \\ \vdots & t=1 \dots n & t=1 \dots n \\ \vdots & & \\ 1 & & \end{bmatrix}_{n \times 3}$$

sm-OLS(y, X_f)

Terminology about Sinusoids

$$s(t) = \boxed{\beta_0} + \boxed{R} \cos(2\pi ft + \phi)$$

- ① β_0 : level of the sinusoid
- ② R : Amplitude (height of maximum oscillation of the sinusoid from β_0)
- ③ f : frequency $\rightarrow$ number of oscillations in unit time.

$\rightarrow$ Hz (Hertz)

$\rightarrow$ number of oscillations in 1 sec.

- ④ period : $\frac{1}{f} \rightarrow$ length of time to complete one oscillation.

- ⑤ ϕ : Phase

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$s(t) = \beta_0 + R \cos(2\pi f t + \phi) \leftarrow$$

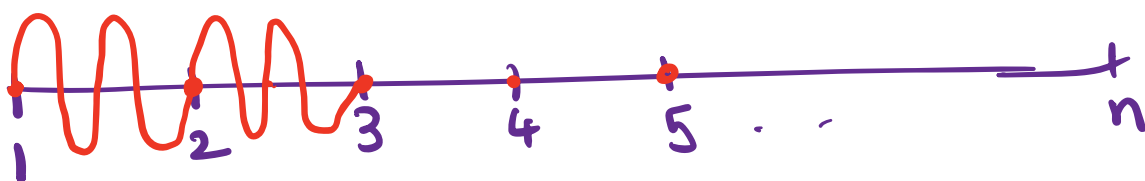
$$= \beta_0 + (R \cos \phi) \cos 2\pi f t + (-R \sin \phi) \sin 2\pi f t$$

$$= \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t$$

$$\beta_1 = R \cos \phi, \quad \beta_2 = -R \sin \phi$$

Aliasing (Discrete Sampling)

$$y_t \quad t = 1, 2, \dots, n.$$



FACT: For every $f \in (-\infty, \infty)$ & $\phi \in (-\infty, \infty)$
 there exist $f_0 \in [0, \frac{1}{2}]$ & $\phi_0 \in (-\infty, \infty)$
 such that
 $\beta_0 + R \cos(2\pi f t + \phi) = \beta_0 + R \cos(2\pi f_0 t + \phi_0)$
 for all $t = 1, 2, \dots, n$

$$f = 2.3$$

$$R \cos(2\pi(2.3)t + \phi)$$

$$= R \cos(\underbrace{2\pi(2)}_{} t + 2\pi(0.3)t + \phi)$$

$$= R \cos(2\pi(0.3)t + \phi)$$

$$f = 0.7$$

$$R \cos(2\pi(0.7)t + \phi)$$

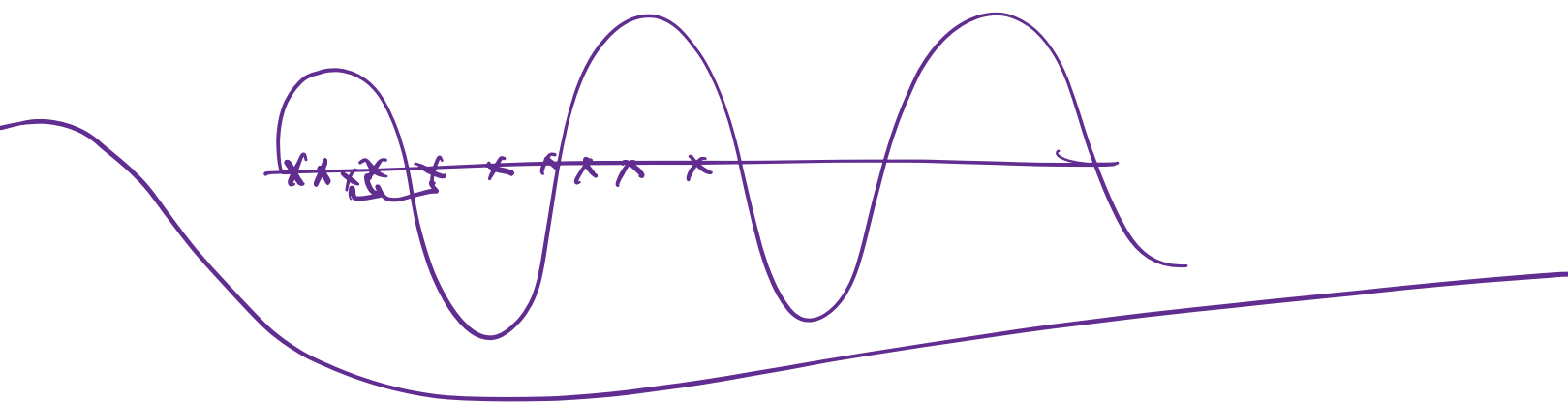
$$= R \cos(\underbrace{2\pi t - 2\pi(0.3)t + \phi}$$

$$= R \cos(2\pi(0.3)t - \phi)$$

$$\cos(2\pi - \alpha) = \cos \alpha$$

For every frequency f , there exist an 'equivalent' frequency $f_0 \in [0, \frac{1}{2}]$

↓
ALIAS of f



$$y = X_f \beta + \varepsilon$$

$$X_f = \begin{pmatrix} 1 & \cos 2\pi f(1) & \sin 2\pi f(1) \\ \vdots & \vdots & \vdots \\ 1 & \cos 2\pi f(n) & \sin 2\pi f(n) \end{pmatrix}$$

Qn 1: How to estimate $f, \beta_0, \beta_1, \beta_2, \sigma$?

① First fix a grid of values of f .

$$[0, \frac{1}{2}]$$

e.g. 0, 0.0001, 0.0002, ..., 0.5

- ② For each grid point f ,
 $\text{sm-OLS}(y, X_f)$ (Linear Regression)

$\text{RSS}(f)$ = residual sum of squares
 in $\text{sm-OLS}(y, X_f)$

- ③ $\hat{f}$ = grid minimizer of $\text{RSS}(f)$

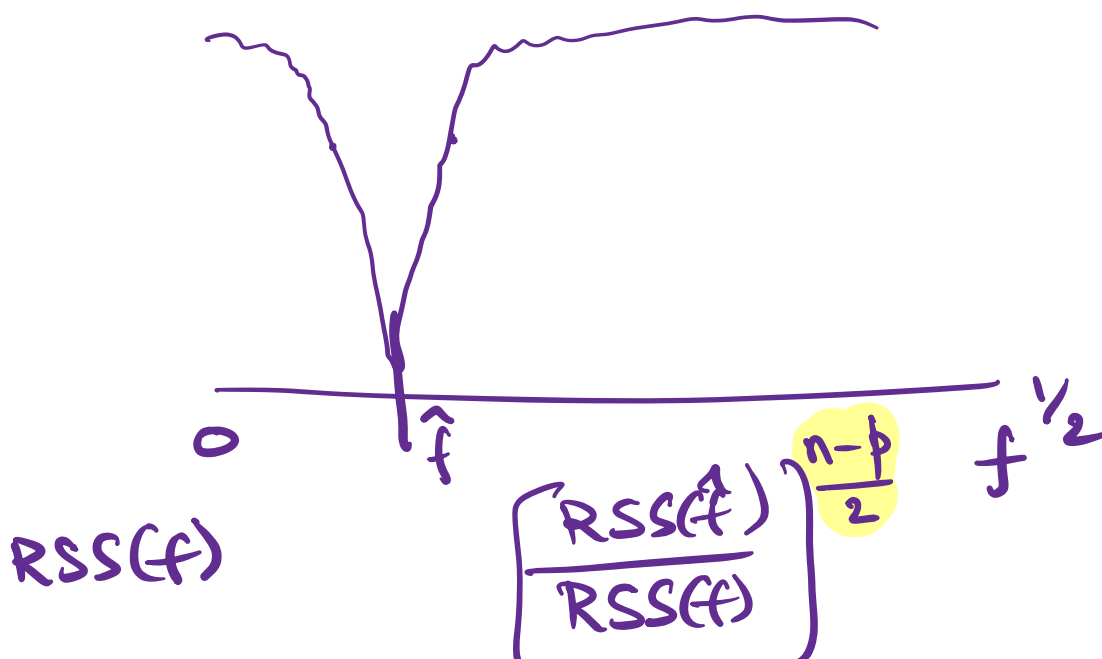
- ④ $\hat{\beta}, \hat{\sigma}^2 \rightarrow$ estimate from OLS of
 y on $X_{\hat{f}}$

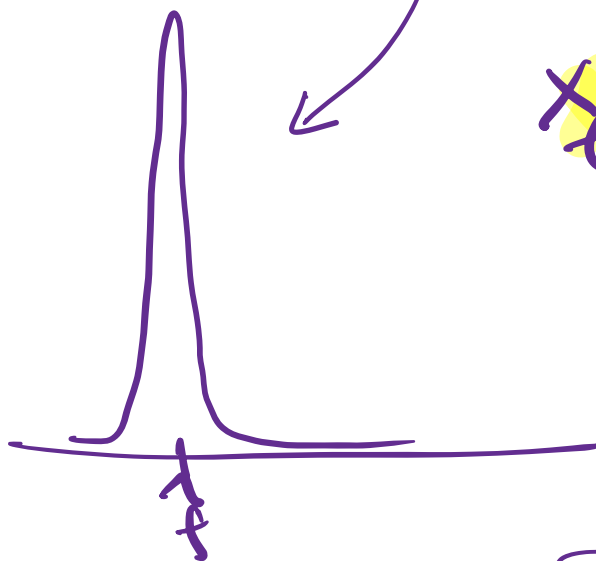
For uncertainty quantification:

$$\text{posterior}(f) \propto |X_{\hat{f}}^T X_f|^{-1/2} \left[\frac{\text{RSS}(\hat{f})}{\text{RSS}(f)} \right]^{\frac{n-p}{2}} \mathbb{I}(0 < f < \frac{1}{2})$$

$\hat{f}$: least squares estimator

$$p=3$$





$$X_t = \begin{pmatrix} \cos 2\pi f t & \sin 2\pi f t \\ t=1 \dots n & t=1 \dots n \end{pmatrix}$$

$$f \in [0, \frac{1}{2}]$$

$$f = 0 \text{ or } \frac{1}{2}$$

$$s(t) = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t$$

$t = 1 \dots n$

$$f = \frac{1}{2}$$

$$\beta_0 + \beta_1 (-1)^t$$

$$\beta_2 \sim \text{Unif}(c, c)$$

